

Cost-Benefit Trade-off Matrix

Value vs. total risk-adjusted cost

PROFESSIONAL DEFINITION

The Cost-Benefit Trade-off Matrix arrays AI initiatives by their expected value against their total, risk-adjusted cost, supporting disciplined prioritisation of where to invest.

WHO DEVELOPED IT

A general decision-analysis construct adapted to AI investment appraisal.

WHY IT WAS DEVELOPED

It was developed to ensure AI investment decisions weigh full risk-adjusted cost against realistic benefit, not headline capability.

FIGURE 1 · COST-BENEFIT TRADE-OFF MATRIX

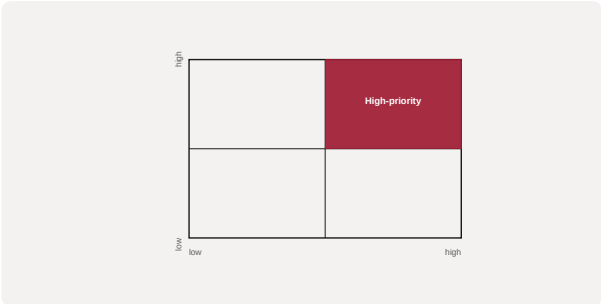


Figure 1. A schematic of the Cost-Benefit Trade-off Matrix as applied in BARBM312 (illustrative).

HOW IT IS USED

Each initiative is scored on value and total risk-adjusted cost; the matrix highlights high-value, justified-cost initiatives.

WHEN IT IS USED

During AI portfolio prioritisation and investment governance.

BY WHOM IT IS USED

Executives, finance and AI programme leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It integrates the cost lessons of the unit — hidden costs and risk — into a single prioritisation view.

RELATED FRAMEWORKS IN THIS COURSE

AI Risk Register · Slibrary 12

Bias & Failure Modes Map · Slibrary 12

Black Swan Radar · Slibrary 12

Mitigation Cost Curve · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

Boardman, A.E. et al. (2017) Cost-Benefit Analysis: Concepts and Practice. Cambridge: CUP.

McKinsey & Company (2023) 'The economic potential of generative AI'.